

Reversal of protein chemical aging by enzymatic deglycation

Received: 16 October 2025

Accepted: 23 June 2026

Published online: 14 July 2026

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The accumulation of advanced glycation end products (AGEs) in long-lived proteins is a hallmark of mammalian aging and implicated as a driver of metabolic dysfunction. Among these adducts, *N*^ε-carboxymethyl-lysine (CML) is particularly abundant in aging tissues, where it modifies proteins and acts as a ligand for the receptor for advanced glycation end products (RAGE), thereby perpetuating chronic inflammation and oxidative stress. While endogenous detoxification systems exist for reactive precursors, the stable CML adduct has historically been considered irreversible. Here, we report the development of CMLase - an enzyme engineered through the directed evolution of over 500 million variants to specifically oxidize CML and restore the native lysine residue. We demonstrate that CMLase effectively reverses CML modifications in model proteins in vitro and in human tissue samples from elderly donors, providing proof-of-concept that protein damage previously deemed irreversible is amenable to enzymatic repair. Collectively, our approach establishes a platform for developing enzymes to reverse age-related molecular damage and ultimately repair tissue proteins compromised by aging and disease.

A defining feature of mammalian aging is the persistent accumulation of chemical damage on proteins of the extracellular matrix (ECM)¹. Advanced glycation and lipoxidation end products (AGEs), formed by the nonenzymatic reaction of sugar- and lipid-derived reactive carbonyls with amino acids, stand out for their abundance and pathogenic effects^{2–4}. Substantial evidence indicates that these abundant adducts are not merely passive biomarkers but active mediators of age-related morbidities^{2–4}. For example, AGE accumulation contributes to the stiffening and loss of elasticity observed in aging tissues by cross-linking collagen and inhibiting normal protein turnover^{5,6}. Moreover, AGEs promote chronic inflammation by acting as neopeptides that engage the adaptive immune system⁷ and by activating the innate immune system via the Receptor for Advanced Glycation End Products (RAGE)⁸.

Among the heterogeneous group of AGEs, *N*^ε-carboxymethyl-lysine (CML) has emerged as a critical, chemically stable adduct found abundantly in long-lived proteins during aging^{2,9,10}. Beyond structural compromise, driven by the conversion of cationic lysine residues to

anionic carboxymethyl adducts, CML exerts deleterious effects through cellular signaling, serving as a specific ligand for RAGE^{11,12}. The engagement of the CML-RAGE axis triggers a signaling cascade that activates NF-κB and stimulates the release of pro-inflammatory cytokines and profibrotic growth factors^{13–16}. In the context of the central nervous system, CML accumulation has been linked to oxidative stress and mitochondrial damage in microglia, further disrupting brain homeostasis during aging¹³. Despite the association between CML accumulation and tissue dysfunction, therapeutic strategies to reverse this modification remain elusive.

A long-standing goal in the fields of diabetes, vascular biology, and aging has been to slow or reverse the accumulation of AGEs in the body^{17,18}. Endogenous detoxification networks, such as the glyoxalase system (Glo1), exist to scavenge reactive dicarbonyl precursors like methylglyoxal, but they do not reverse stable AGE adducts once formed on proteins¹⁹. Pharmacological interventions, such as aminoguanidine and alagebrium, have demonstrated the ability to inhibit formation by trapping or breaking up reactive intermediates²⁰. While

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this strategy can reduce the formation of new AGEs, it does not address the substantial pool of pre-existing AGEs that have accumulated over decades and these approaches do not restore the native protein structure.

Here we describe the development of CMLase, an enzyme engineered to specifically reverse CML modifications formed on protein substrates. Through computational screening and directed evolution, we engineered glycine oxidase to oxidize CML and restore the native lysine. We show that CMLase reverses CML modifications on model proteins as well as in aged human tissues known to accumulate substantial AGE burden. This work establishes that damage to aging proteins previously thought to be irreversible can be enzymatically repaired, providing a foundation for developing interventions that target the molecular underpinnings of human aging.

Results

Structural mining identifies a glycine oxidase scaffold accessible to peptide substrates

A CML-cleaving enzyme with activity on peptides and proteins is not reported in the literature. To engineer a biocatalyst capable of this repair, we sought to identify a suitable starting point that cleaves either free CML or peptidyl-CML. We reasoned that a sufficiently active starting point could be enhanced through application of directed evolution to yield a highly active protein-CML cleaving enzyme. We first examined the literature for reports of enzymatic activity on CML. One study partially isolated and characterized a flavoenzyme with CML oxidase activity in cell lysate from a *Cladosporium* soil mold²¹. Unfortunately, we were unable to identify the reported enzyme as no DNA or protein sequence information was reported, the original *Cladosporium* isolate was unavailable (personal communication), and we found that cell lysate from a commercial *Cladosporium* isolate was inactive on CML and peptidyl-CML. Therefore, we sought to identify an alternative starting point.

We identified the FAD-dependent enzyme glycine oxidase as a candidate for CML cleavage based on chemical similarity between the carboxymethyl-amino moiety of CML and glycine²² (Fig. 1A). A glycine oxidase candidate from *Bacillus subtilis* was reported to have activity on a variety of *N*-alkyl derivatives of glycine, suggesting the active site might accommodate bulkier substrates such as CML²³. We recombinantly produced *B. subtilis* glycine oxidase (BsGO) and observed activity on free CML with lysine, glyoxylic acid, and hydrogen peroxide produced as products (Fig. 1A, B). BsGO exhibited promiscuous activity, showing a catalytic efficiency on CML ($k_{\text{cat}}/K_M = 8.6 \times 10^{-3} \text{ s}^{-1} \text{ mM}^{-1}$) that was approximately thirty-fold lower than on its native substrate, glycine ($k_{\text{cat}}/K_M = 0.24 \text{ s}^{-1} \text{ mM}^{-1}$) (Supplementary Table 1). However, BsGO showed no detectable activity when assayed against peptidyl-CML substrates. This result prompted us to seek alternative starting points that either possessed initial activity against peptidyl-CML or demonstrated higher initial activity against free CML.

We initiated the screening of BsGO homologs to identify a better starting point for enzyme engineering. We chose sixteen sequences to evaluate against CML and peptidyl-CML. Of these, thirteen BsGO homologs were successfully produced in soluble form. Notably, eight of these showed measurable activity on free CML, which suggests either considerable substrate promiscuity or that some enzymes annotated as glycine oxidases may be misannotated (Fig. 1B). None of the enzymes had detectable activity on peptidyl-CML. Notably, the glycine oxidase from *Bacillus* sp. INLA3E (BsIGO) displayed a three-fold lower K_M value for CML (2.6 mM) compared to BsGO (8.2 mM) (Fig. 1D). Comparison of the crystal structure of BsGO with an AlphaFold model of BsIGO²⁴ showed that both enzymes have a similar active site; however, key differences such as His²⁴⁴, BsGO $\rightarrow$ Gly²⁴⁴, BsIGO show a trend toward smaller residues in the active site. We reasoned that the presence of smaller amino acid side chains in the BsIGO active site could allow for better accommodation of CML

compared to the BsGO active site (Supplementary Fig. 1). Modeling of CML into the active site of BsIGO revealed structural features of the enzyme that were likely blocking access of CML peptide substrates to the active site (Fig. 1C).

Given the high frequency of promiscuous CML activity among annotated glycine oxidases, we sought a GO homolog with minimal secondary structure that would obstruct access to the active site (Fig. 1C). Using Uniprot, we selected all sequences annotated as either a glycine oxidase or an amino acid oxidase that also had an available AlphaFoldDB structure²⁵. This search yielded 44,783 sequences. We then performed pairwise structural alignments to BsIGO, specifically screening for enzymes exhibiting a deletion of ten or more amino acids in the $\alpha 9$ helix region (Fig. 1C). Fewer than fifty unique sequences were identified that met this criterion and we experimentally identified a glycine oxidase from *Calidithermus roseus* (CrGO) which displayed detectable activity on a peptidyl-CML substrate (AA-CML-AA). CrGO, when compared to BsIGO, featured a twenty amino acid deletion near the active site. This deletion included the complete absence of the twelve amino acid helix ($\alpha 9$), which we reasoned facilitates access to larger substrates (Fig. 1C). Due to CrGO's notably low activity on peptidyl-CML ($k_{\text{cat}}/K_M = 1.2 \times 10^{-3} \text{ s}^{-1} \text{ mM}^{-1}$), we aimed to develop an engineering strategy to enhance its activity toward peptide substrates.

A genetic selection enables the engineering of a highly active protein CMLase

To increase the peptidyl-CML oxidase activity of CrGO, we developed a high-throughput genetic selection strategy. Genetic selections enable rapid screening of large DNA libraries ($>10^8$ members) to identify rare enzyme variants with desired activities which offers advantages over more limited screening-based approaches²⁶. We first developed a genetic selection for CML oxidase activity by coupling the growth of an *E. coli* lysine auxotroph to the activity of CML oxidase, which produces lysine from free CML. We then adapted the selection on free CML to develop a selection to identify enzyme variants with activity on peptidyl-CML (Fig. 2A). The peptidyl-CML selection used a peptide substrate small enough (seven amino acids) to enter the periplasm but too large to be transported into the cytoplasm, ensuring that the peptide was not subject to degradation by intracellular peptidases. CML oxidase variants were fused to an N-terminal signal peptide to localize the enzyme in the periplasm where active variants convert peptidyl-CML into peptidyl-lysine. Subsequently, trypsin localized to the periplasm cleaves after the newly formed lysine residue, generating smaller peptide fragments that are transported into the cytoplasm. The small peptides are then proteolyzed in the cell which enables growth of the lysine auxotroph. We validated the selection by showing that growth on peptidyl-CML was dependent on the presence of a CML oxidase with activity on peptidyl-CML (Fig. 2B). With a functional selection in place, we next sought to evolve CrGO into a highly active enzyme for cleaving CML modifications on peptides and proteins.

We first focused on improving activity of the enzyme on free CML by picking the fastest growing colonies from the selection and assessing hits for improved activity on CML and peptidyl-CML. In the first round of evolution, we constructed a variant library using error-prone PCR. The library was evaluated for activity in both the free CML and peptidyl-CML selection formats. While no variants demonstrated activity in the peptide selection, even under permissive conditions (1 mM peptide and seven days of growth), selecting on free CML yielded a lawn of colonies. To isolate the most improved variants, we increased the selection stringency by reducing IPTG (20 μM) and CML (100 μM) concentrations and only selecting colonies that appeared within two days. This process yielded the variant CrGO-764 which harbored seven amino acid mutations (A35T, P87S, V129A, R141C, M194I, F195C, G243R) and exhibited a fivefold improvement in activity on peptidyl-CML (Supplementary Table 2).

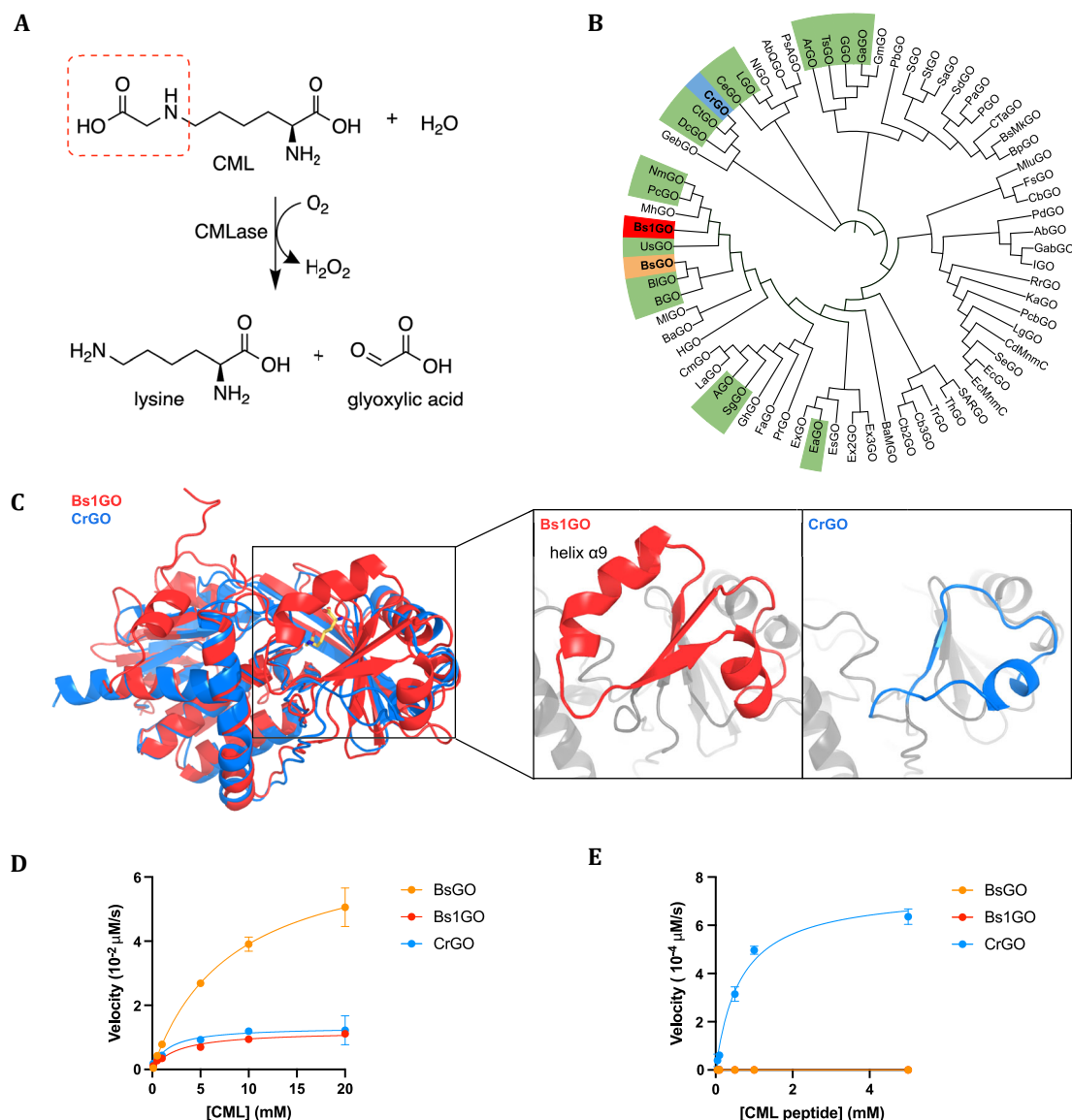


Fig. 1 | Discovery of glycine oxidase homologs with activity on peptidyl-CML.

A CML oxidase reaction catalyzes oxidation of the epsilon nitrogen of CML forming lysine with hydrogen peroxide and glyoxylic acid as byproducts. A box is drawn around the substructure glycine moiety within CML. **B** Various glycine oxidase homologs were tested for activity on CML (proteins harboring CML oxidase activity highlighted in green). Sequences were identified by sequence similarity search using BsGO (orange) or Bs1GO (red) as queries in the databases UniProt and NCBI. Structural searches led to the identification of CrGO (blue). Phylogenetic tree was generated in ClustalW alignment, visualized in iTOL. **C** Overlay of AlphaFold models

of Bs1GO (red) and CrGO (blue) with CML docked into the active site (yellow). Cutout displays helix α9 in Bs1GO, conserved across the majority of enzymes tested, and the deletion of helix α9 in CrGO. **D** Michaelis-Menten plots of CML oxidase enzymes on CML and **E** CML peptide substrates (AA-CML-AA). Kinetics were measured in triplicate by continuous assay, coupling H₂O₂ byproduct of the oxidation reaction to a fluorometric readout using horseradish peroxidase. BsGO and Bs1GO had no detectable activity on the peptide substrate. Data are presented as mean values ± SD, *n* = 3.

Analysis of a homology model of CrGO-764 revealed two unstructured loops (Loops 1 and 2) partially obstructing the active site opening and potentially hindering substrate access (Fig. 2C, Supplementary Fig. 2). Therefore, in the second round of evolution, we targeted these loops. Libraries were generated to randomize sequences in each loop and introduce deletions of one to four amino acids. Growth selection indicated that deletions in loop 1, specifically one and two amino acid reductions, yielded functional variants under permissive free CML conditions. Attempts at larger deletions of loop 1 or modifications to loop 2 resulted in non-functional enzymes. A highly active variant was identified (CrGO-785) which contained an amino acid substitution (H49W) and a two amino acid deletion (Ala⁵⁰ and Glu⁵¹) in loop 1 (Fig. 2C). CrGO-785 showed both improved affinity for peptidyl-CML and greatly improved growth in the selection (Supplementary Table 3).

Next, we made active site targeted libraries of CrGO-785 guided by a GO homolog from *Candidatus Eremiobacteraeota* (CeGO) that displayed much higher affinity for free CML ($K_{M,CeGO}$ = 0.059 mM, $K_{M,CrGO-785}$ = 0.94 mM). We reasoned that if we could mimic the residues deep in the active site which strongly interact with the CML side chain, without changing the residues on the surface that might restrict access of larger peptide substrates, we could increase CrGO-785's affinity for CML-peptide substrates. Structural comparison of CeGO and CrGO revealed high conservation of most first shell amino acids in the substrate binding pocket with five key differences; Arg^{60,CeGO} → Ala^{61,CrGO}, Asp^{143,CeGO} → Gly^{112,CrGO}, Arg^{232,CeGO} → Cys^{195,CrGO}, Asp^{235,CeGO} → Ala^{198,CrGO}, and Glu^{252,CeGO} → Gln^{214,CrGO} (Fig. 2D). We simultaneously randomized all five sites on CrGO-785 and used the free CML genetic selection to isolate the most active variants. Hits from the selection

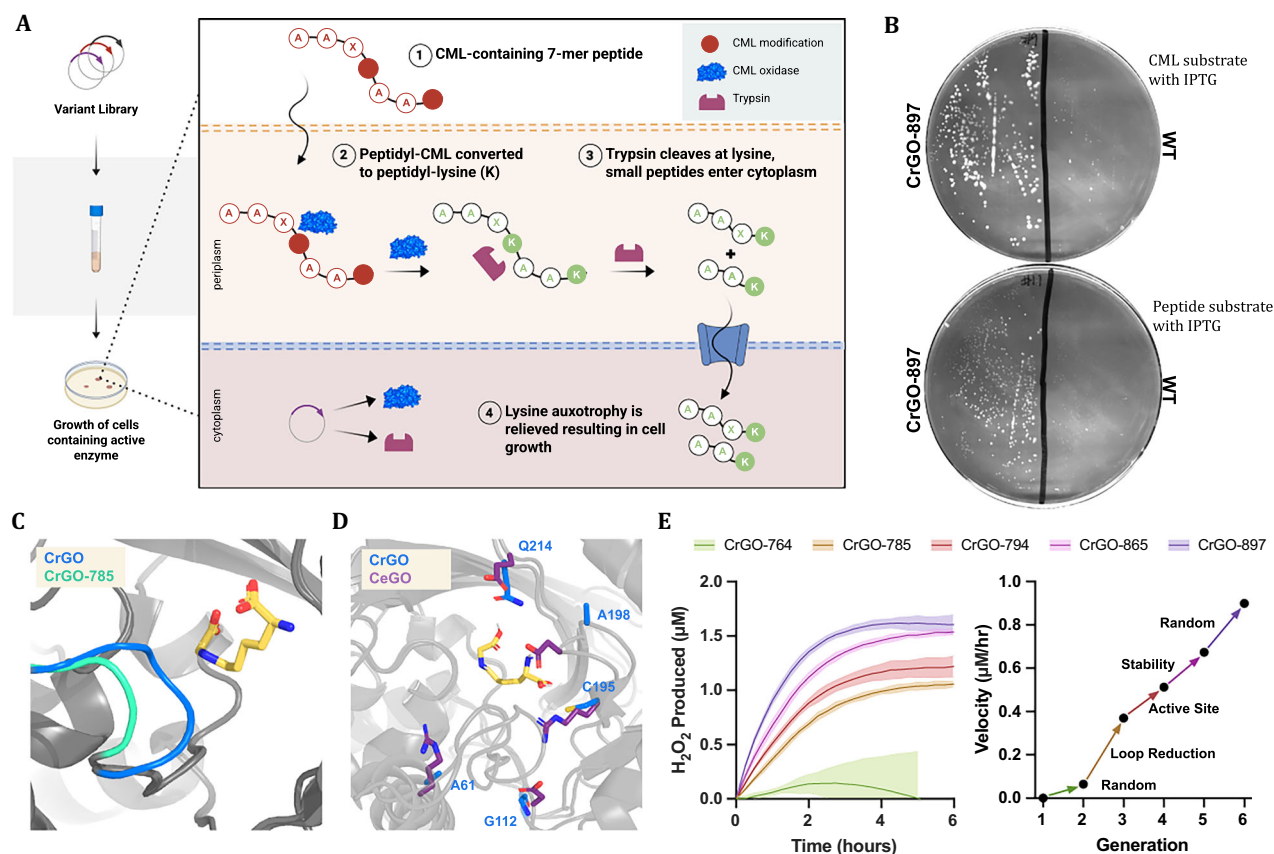


Fig. 2 | Engineering a highly active peptidyl-CML oxidase. A Biochemical logic describing the genetic selection used to engineer CML oxidase. A DNA library of periplasmically localized CML oxidase variants is transformed into a lysine auxotroph strain of *E. coli*. A CML-containing heptapeptide substrate can access the periplasm but is unable to enter the cytoplasm due to size limits of inner membrane peptide transporters. Cells producing an active peptidyl-CML oxidase variant convert peptidyl-CML into peptidyl-lysine which is subsequently cleaved by periplasmically localized trypsin. The smaller peptide fragments enter the cytoplasm and relieve the lysine auxotrophy, enabling cell growth. **B** Growth of a lysine auxotroph is dependent upon the presence of CML or CML peptide and a CML oxidase with activity on free or peptidyl-CML. Plates are of lysine auxotrophs grown on free CML or CML peptide, co-expressing trypsin and the noted CML oxidase variant (WT, CrGO-897). The engineered variant (CrGO-897) grows significantly better on

both free CML and CML peptide. **C** AlphaFold models of WT CrGO (blue) and CrGO-785 (cyan). Loop 1 of CrGO-785 has been reduced by two amino acids, causing a noticeable decrease in loop size in the structure. **D** AlphaFold models of WT CrGO (blue) and the homolog with the highest free CML activity, CeGO (purple), in complex with free CML (yellow). Active sites are similar except for the highlighted five residues which were targeted for libraries. **E** Five rounds of engineering yielded significant improvement in CML oxidase activity (5 μ M) on CML-BSA (1 μ M). A continuous assay detecting H_2O_2 was used to assess CML oxidase variants with the lines representing the average of three trials. Results are shown over time with background signal from unmodified BSA removed. Initial velocity on CML-BSA was calculated over the first hour. Arrows indicate the libraries that resulted in each variant. Data are presented as mean values $\pm$ SD, $n = 3$.

revealed a consensus set of mutations on the CrGO-785 parent at three sites (A61G, A198S, and Q214K). The new variant, CrGO-794, displayed a three-fold higher catalytic efficiency on free CML ($k_{cat}/K_M = 0.023 \text{ s}^{-1} \text{ mM}^{-1}$) and a 50% improvement in activity on peptidyl-CML (Supplementary Table 3).

We observed that some hits from the free CML selection failed to display improvements in enzyme activity. We reasoned that the amino acid substitutions in these hits conferred improvements in enzyme stability that provided an advantage in the selection. To test this hypothesis, putative enzyme stabilizing variants were combined as single or double amino acid substitutions on top of CrGO-794. We characterized the thermostability and peptide activity of these enzymes and identified a variant (CrGO-865) with two substitutions adjacent to the active site (L83I, S85T) which resulted in improvements in thermostability and catalytic efficiency (Supplementary Table 3).

We then sought to engineer increased peptidyl-CML activity into CrGO-865 by constructing an error-prone PCR library and identifying improved variants in the peptidyl-CML selection. To prevent enzyme bias toward specific peptide sequences, we challenged the library for growth on mixtures of poly-alanine peptides containing seventeen

different amino acids (X) adjacent to the internal CML modification (AAX-CML-AA-CML and AA-CML-XAA-CML). Under these conditions, only cells expressing CML oxidase variants capable of oxidizing CML and tolerating variation in the residues flanking the CML site survived. From this selection, we identified the variant CrGO-897, which harbors three mutations (E80Q, H226A, H230R) and exhibits a 20% improvement in catalytic efficiency on peptide (Supplementary Table 3). CrGO-897 retained high specificity for the carboxymethyl-amino moiety of glycine and CML: it did not oxidize other canonical amino acids (Supplementary Fig. 5), did not oxidize glycine or arginine positioned at the termini of tripeptide substrates (Supplementary Fig. 6), and showed no activity on carboxymethyl-arginine (CMA) protein substrates (Supplementary Fig. 7). In summary, after five rounds of directed evolution, we developed a highly active CMLase (CrGO-897) featuring fifteen amino acid substitutions and a two amino acid deletion compared to the original CrGO enzyme.

We next evaluated CMLase activity on CML-modified bovine serum albumin (CML-BSA), a commonly used AGE carrier protein. CML-BSA was prepared using established protocols²⁷ and we demonstrated that CMLase catalyzes the specific oxidation of protein-bound

CML modifications. Incubation of CMLase (5 μ M) with CML-BSA (1 μ M protein) produced hydrogen peroxide at a rate of 0.90 μ M/h (Fig. 2E). No hydrogen peroxide was detected in control reactions lacking CMLase or when the enzyme was incubated with unmodified BSA (Supplementary Fig. 8). To confirm that the observed deglycation reflected CML removal from intact protein rather than proteolytic degradation, we analyzed the reaction products by SDS-PAGE and western blot. SDS-PAGE showed that the protein remained intact following overnight incubation with CMLase, with no evidence of fragmentation (Supplementary Fig. 9), and western blot using an anti-CML antibody showed an enzyme-dependent reduction in CML-BSA signal (Supplementary Fig. 10). Remarkably, each round of enzyme evolution yielded improvements in activity on CML-BSA, despite selections being conducted using only free CML and peptidyl-CML substrates (Fig. 2E).

CMLase Effectively Reverses CML Modifications on a Wide Range of Protein Scaffolds

Physiologically, CML occurs at solvent-exposed lysine residues of long-lived ECM proteins, particularly in the basement membrane of tissues like the eye²⁸, skin², kidney, and vascular wall^{29,30} where it disrupts ECM-cell signaling and promotes inflammation via interactions with RAGE¹². To assess the potential of CMLase to accept physiologically relevant protein substrates, we next evaluated its *in vitro* activity on a panel of physiologically relevant proteins including casein, hemoglobin, collagen, and a retinal total protein extract (eye TPE) from sheep. A CML ELISA assay was used to measure CML levels in each sample (Supplementary Fig. 11) and the specificity of the anti-CML antibody was confirmed by comparing CML-BSA with unmodified BSA. Treatment of CML-modified proteins with CMLase (5 μ M, overnight) resulted in significant reductions of CML ELISA signals that ranged from 52% to 97% (Fig. 3A). These data demonstrate that CMLase effectively removes the majority of CML modifications on the protein substrates that are detectable by ELISA. The variation in the amount of CML removal between protein substrates is likely due to variability in lysine side-chain accessibility and local structural context. To confirm that the CML removal detected by ELISA is indeed due to CMLase catalytic activity, we included a catalytically inactive enzyme control (CeGO-763) which showed no significant reduction in CML ELISA signal. Importantly, these ELISA results were reproducible across varying degrees of CML modification (Supplementary Table 4), reinforcing the robustness of our data.

We next evaluated the site-specific repair activity of CMLase using proteomics to map the occurrence of CML modifications on BSA. CMLase was able to act on the majority of modified sites: of thirty-three CML modified lysine residues detected by proteomics, thirty showed reduced CML levels after CMLase treatment (Fig. 3B, C). Notably, twenty-one sites exhibited reductions greater than 50%, with seven sites showing reductions of more than 90% (Fig. 3B, Supplementary Fig. 12). CMLase activity on a particular site was not strictly a function of surface accessibility (Fig. 3C). For example, lysines K51, K232, and K573 all reside in surface-exposed helices but showed varied CML reductions of 90%, 38%, and 0%, respectively. Activity weakly correlated with local protein flexibility (B-factor), suggesting flexibility may be necessary but is not sufficient for catalysis (Supplementary Fig. 13). Some local sequence preference was observed, with CMLase showing higher activity on sites with neighboring cysteines and reduced activity on sites followed by hydrophobic regions (Supplementary Fig. 14). These trends suggest that CMLase may favor more hydrophilic, solvent accessible sites within the BSA structural environment. While most surface-exposed sites were reduced, three heavily modified lysines (K131, K504, K573) showed less than 5% reduction, likely contributing to residual ELISA signal observed in CMLase treated CML-BSA (Fig. 3A). Reduced enzymatic activity was associated with bulky hydrophobic or

aromatic residues (e.g., F, W, L, V) in the immediate flanking region—a potential indicator of reduced accessibility, exemplified by resistant sites such as K154, K528, and K573 (Supplementary Fig. 14). Importantly, while CMLase acts broadly across most modified sites, certain AGE modifications are expected to be more pathogenic depending on the protein and disease context. Therefore, targeting key “hot-spot” residues rather than complete modification removal may suffice to restore protein function^{14,31–33}. Future engineering efforts could refine CMLase specificity to selectively focus on these critical sites. Together, these findings demonstrate that CMLase efficiently reverses CML modifications on a range of physiologically relevant proteins and is active across the majority of surface-exposed sites.

CMLase reverses endogenous CML accumulation in aged human tissues

We then sought to test the ability of CMLase to reverse endogenously formed CML modifications. CML accumulation occurs predominantly in long-lived organisms making short-lived animal models ill-suited for studying the buildup of this AGE. We therefore tested CMLase directly on human tissues where CML has had decades to accumulate. Here, we show that CMLase reverses the majority of endogenously produced CML modifications in human lens, skin, and arterial tissue.

Lens crystallins are among the longest-lived proteins in the human body which makes them susceptible to CML accumulation. Lens tissue from a 64-year-old donor was homogenized and soluble protein fractions were incubated overnight with CMLase (5 μ M). Proteomics evaluation of the lens crystallins proved challenging due to protein heterogeneity; therefore, bulk CML content was assessed by hydrolyzing the samples and measuring CML by a standard addition method LC-MS/MS³⁴. Remarkably, CMLase reduced the total concentration of CML modifications in lens proteins by 45% compared to untreated control samples (Fig. 4A). The reduction of CML content was validated by a CML ELISA which showed a 78% reduction in the total CML content in samples treated with CMLase compared to untreated control samples (Fig. 4B). The difference in CML reduction by ELISA (78%) compared to LC-MS/MS (45%) is consistent with surface-exposed immunologically detectable CML being more accessible to CMLase than the total CML pool, which may include buried residues.

We next examined the ability of CMLase to reverse physiologically accumulated CML directly on intact human tissues. Immunohistochemical staining (IHC) for CML was performed on formalin-fixed, paraffin-embedded sections of human abdominal aorta and skin from donors aged 20 to 75-years-old. Antigen retrieval steps were omitted to avoid heat-induced *de novo* formation of CML modifications²⁹. Several commercial anti-CML antibodies were evaluated for their staining performance and specificity. While all tested antibodies displayed reactivity, competitive inhibition with CML-BSA antigen confirmed the monoclonal anti-CML antibody 6D12 (Cosmo Bio) as highly specific for CML (Fig. 4C). Consequently, 6D12 was used for all subsequent IHC experiments. Consistent with the literature, we observed age-dependent IHC staining of CML in human arterial tissue²⁹. Arterial tissue from donors 20 to 25-years-old lacked specific staining of CML modifications with the anti-CML antibody (Supplementary Fig. 15); whereas arterial tissue from a 75-year-old donor showed strong staining, especially along the arterial wall and adjacent to atherosclerotic lesions (Fig. 4C, Supplementary Fig. 17). Thus, our IHC protocol displays specificity for CML and shows an expected increase in CML staining in aged human tissue.

We then assessed the ability of CMLase to reverse CML in arteries and skin by incubating tissue sections overnight with CMLase (5 μ M) or an inactivated control enzyme (CeGO-763), followed by IHC staining of CML modifications. All CMLase treated sections showed a visually apparent reduction in CML-specific staining following CMLase treatment (Fig. 4D, Supplementary

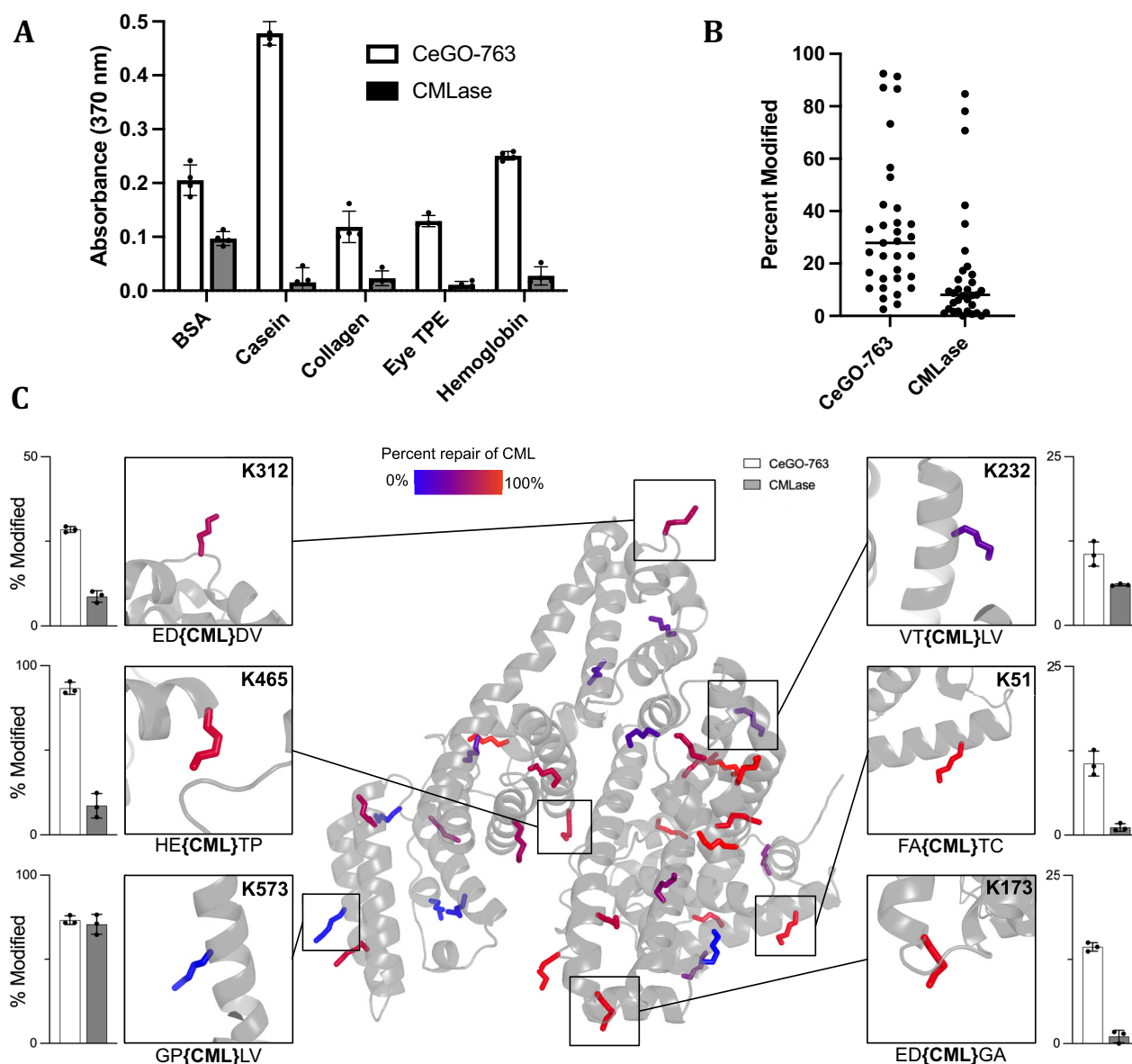


Fig. 3 | CMLase reverses a majority of CML damage on proteins. **A** CMLase is active on a range of CML-modified protein substrates. Proteins were bound to a 96-well plate and CML modifications were introduced using the glyoxylic acid (GA) method. GA concentrations were as follows: BSA (300 μ M), collagen (100 μ M), hemoglobin (50 μ M), casein (300 μ M) and eye total protein extract (eye TPE, 100 μ M) and were selected to achieve similar degrees of modification between the various proteins. A comprehensive set of conditions are shown in Supplementary Table 4. CML-modified proteins were treated with CMLase or an inactive control overnight. ELISA data shown is absorbance measured at 370 nm with background signal from control samples unmodified by GA subtracted. $n = 4$. **B** CMLase generally reduces CML levels in CML-BSA across 33 sites. Proteomics was used to map

the site specific changes in CML on CML-BSA following treatment with CMLase or an inactive control (CeGO-763). Each data point represents the percent of lysines with CML modifications at a single site when treated with each enzyme. **C** CMLase has varying activity across all sites with no clear preference based on primary or secondary structure. All 33 analyzed lysine sites are colored based on percent repair of CML into lysine. A selection of lysine sites is highlighted on BSA (PDB 4F5S) in a variety of structural contexts, chosen to highlight structural diversity. Protein sequence context is also provided for each site. Bar graphs indicate the percent of lysine residues that are modified to CML at a specific site with inactive enzyme (white) or CMLase (gray) treatment. Data are presented as mean values $\pm$ SD, $n = 3$.

Fig. 16). CML staining was quantified by analyzing DAB intensity across four spatially aligned subregions and showed that CMLase treatment reduced CML content by more than 70% in elderly arterial tissue (Fig. 4E, Supplementary Fig. 17). In elderly human skin, CMLase treatment resulted in more than a 55% reduction in CML staining in epidermal and dermal layers and reversed levels of CML staining to less than those found in 31-year-old skin (Supplementary Figs. 18–20). Collectively, these data highlight the potential of CMLase to repair physiologically formed CML damage across diverse tissue types and protein substrates.

Discussion

Our work addresses a long-standing challenge in aging biology by demonstrating the enzymatic reversal of protein chemical aging using an engineered deglycase. CML has historically been categorized as a chemically stable and irreversible advanced glycation end product (AGE)³⁵. As a result, it accumulates on long-lived proteins, disrupting the structural integrity of the extracellular matrix (ECM) and contributing to chronic inflammation through the activation of the receptor for advanced glycation end products (RAGE)¹². While endogenous detoxification systems, such as the glyoxalase pathway, exist to

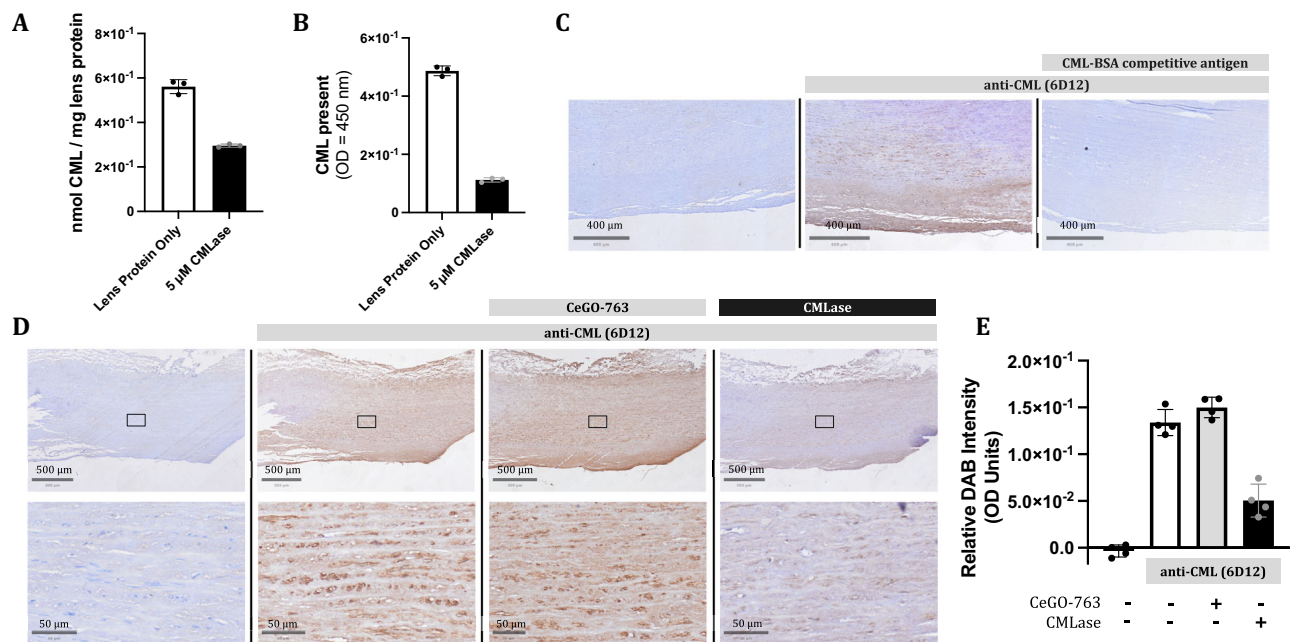


Fig. 4 | CMLase reverses CML modifications in human tissue. **A** CMLase reverses CML modifications in human lens proteins. Soluble human lens proteins were treated with and without CMLase overnight. Samples were subjected to acid hydrolysis and CML content was analyzed by LC-MS/MS multiple reaction monitoring (MRM). Ion counts were normalized by total phenylalanine content to account for differences in hydrolysis efficiency, and quantified with an external standard curve. Normalized and unnormalized values of CML were similar. **B** Soluble human lens proteins were bound to a 96-well plate and treated with and without CMLase overnight. ELISA data shown is absorbance measured at 450 nm with background signal from control samples lacking a primary anti-CML antibody subtracted. **C** Anti-CML antibody 6D12 specifically stains CML modifications in human arterial tissues. Consecutive sections of abdominal aorta from a 75-year-old

human donor stained with H&E and by immunohistochemistry (IHC) using anti-CML antibody 6D12. IHC staining was performed in the absence of 6D12 as a negative control (left), with 6D12 (middle) and with 6D12 in the presence of 2 mg/mL of competitive antigen CML-BSA (right). **D** CMLase treatment reduces CML staining in human arterial tissues. Consecutive sections of abdominal aorta from a 75-year-old human donor incubated overnight with no enzyme (panels 1–2), with an inactivated control (panel 3) or CMLase (panel 4). Following overnight incubation, sections were washed and stained with H&E and by immunohistochemistry (IHC) using anti-CML antibody 6D12 (panels 2–4). **E** Relative DAB Intensity was calculated as optical density (OD) from DAB-stained regions ($n = 4$ regions) using color deconvolution in ImageJ. OD values reflect DAB intensity with higher values indicating stronger staining. Data are presented as mean values $\pm$ SD.

scavenge reactive dicarbonyl precursors like methylglyoxal, they are incapable of reversing stable AGE adducts once formed on proteins³⁶.

To overcome this biological bottleneck, we employed a large-scale computational screen followed by directed evolution to engineer CMLase. The starting scaffold, a glycine oxidase, initially displayed low promiscuous activity on CML. Over five rounds of evolution, we generated the variant CrGO-897 (CMLase), yielding a >10 -fold improvement in catalytic efficiency on peptide substrates and establishing a de novo capacity to deglycate full-length proteins.

While the catalytic efficiency of CMLase on peptide substrates remains approximately 10–50 fold lower than that of evolutionarily perfected PTM editors like Lysine Specific Demethylase 1 (LSD1)³⁷, it represents a significant improvement over the starting scaffold and establishes distinct de novo enzymatic activity on full-length proteins. We observed variability in repair efficacy across substrates (e.g., reductions ranging from 52% (BSA) to 97% (casein) across protein substrates), suggesting a complex interplay between substrate accessibility and antibody detection efficiency. Probing the site-specific nature of CMLase using proteomics on BSA suggests that the residual ELISA signal may be derived from just a few CMLase-resistant immunodominant modifications. Finally, the ability of CMLase to reduce CML burden by over 70% and 55% in elderly human arterial and skin tissue, respectively, demonstrates that the enzyme functions effectively in the heterogeneous and sterically complex environment of the aging human ECM.

Despite these promising ex vivo results, limitations regarding functional restoration and safety must be addressed prior to clinical translation. While we demonstrated the chemical reversal of CML, it

remains to be determined if this translates to the recovery of tissue biomechanics or the silencing of pathogenic RAGE signaling in vivo. Furthermore, our current findings in skin, arterial, and lens tissues utilized homogenized protein or thin, formalin-fixed paraffin-embedded (FFPE) sections where substrate accessibility is maximized. In the context of intact, living organs, the ability of an enzyme to penetrate the dense, cross-linked architecture of the extracellular matrix (ECM) must be evaluated. The substrate scope of CMLase also remains to be fully delineated; while we have shown that the enzyme is selective for the lysine ϵ -amine adduct over the chemically related arginine adduct CMA, activity on α -amine carboxymethylation at protein N-termini was not directly tested and warrants future characterization. Additionally, the enzyme's bacterial origin necessitates a rigorous evaluation of immunogenicity, potentially requiring protein de-immunization or immunomodulation strategies if repeat dosing is required. Regarding reaction byproducts, hydrogen peroxide and glyoxylate are naturally present in biofluids and efficiently cleared by metabolic enzymes^{38,39}. Because CML exists at low concentrations relative to bulk metabolites, and CMLase exhibits minimal off-target activity, the efflux of byproducts generated during repair is expected to be negligible relative to endogenous clearance capacity. Future studies will need to address the pharmacokinetics of tissue penetration and potential immunogenicity, ensuring that CMLase can reach sequestered modifications within the interstitial space to achieve meaningful systemic or localized repair. Ongoing work in our laboratory is focused on further optimizing CMLase catalytic efficiency and adapting the enzyme for the requirements of translational applications.

This work establishes a proof-of-concept for the deglycation of aging tissues. CML is a major ligand for RAGE, and RAGE signaling has been implicated in many age- and diabetes-associated complications^{8,12,40}. Thus, reducing the burden of CML on long-lived, aged proteins may represent a viable strategy to lessen disease burden in aging and diabetic individuals. While CML is only one of many age-related modifications and its reversal alone will not resolve the multifactorial phenotype of aging, the directed evolution platform established here can be adapted to target other pathogenic AGEs. For instance, glucosepane is a dominant cross-link in human tissues that may compromise tissue elasticity and is currently resistant to reversal⁵. The successful engineering of CMLase suggests that other oxidative or hydrolytic enzymes could be evolved to target the chemically diverse landscape of age-related protein modifications. By reversing a hallmark of aging, CMLase provides a powerful tool for dissecting molecular causality and offers a foundation for developing regenerative therapies that repair damaged tissues.

Methods

Statistical methods

Experimental samples were typically analyzed in triplicate, and error bars signify the standard deviation (SD) unless specified otherwise.

Computational screening of glycine oxidases for minimized helix $\alpha 9$

Given the high frequency of promiscuous CML activity among annotated glycine oxidases (GOs), we initiated a campaign to discover enzymes with a substantially reduced helix $\alpha 9$. We first queried UniProt for sequences annotated as “D-amino acid oxidase”, “amino acid oxidase”, or “glycine oxidase” that had corresponding AlphaFoldDB structures available (<https://alphafold.ebi.ac.uk/>). These search terms returned 71,426 non-redundant sequences in UniProtKB, of which 56,499 were found to have associated AlphaFoldDB structures. We further filtered these by length (250–650 amino acids), reducing the set to 44,783 sequences. The 44,873 sequences were screened by pairwise structural alignments to BsIGO and assessed for a ≥ 10 -residue deletion in the helix $\alpha 9$ region. All structural alignments were performed in a PyMOL script using the *align* command with CrGO as the reference. For each query structure, we counted all atoms lying within a 5 Å radius of a pseudo atom positioned near CrGO helix $\alpha 9$ region. The total atom count in this region was used to determine whether helix $\alpha 9$ was absent or significantly truncated. This process yielded ten sequences with substantial deletions and reductions in the helix $\alpha 9$ region. The ten sequences were used as queries for BLAST searches in NCBI and UniProt to identify additional sequences lacking AlphaFold structures but still containing similar deletions. Fewer than fifty unique sequences satisfied these criteria; from these, forty were selected for experimental characterization.

Structural analysis

Protein structures were analyzed and visualized using Pymol Version 3.1.6.1. Enzyme variants without AlphaFold models in the UniProt database were modeled using ColabFold v1.5.5. Free CML was docked into the active site of CrGO using Glide in Bioluminate (Schrödinger).

Random library generation

Agilent's GeneMorph II Random Mutagenesis Kit was used to create random variant libraries. The first PCR reaction was performed using 1 ng of template DNA and 25 cycles. 1 ng of the resulting mutated DNA was then used in a second 25-cycle mutazyme reaction to obtain a higher mutation frequency of 2–8 amino acid mutations per gene. Mutated DNA was then cleaned up using TakaraBio Nucleospin Gel and PCR Clean Up Kit and the resulting DNA was used in a BamHI-HF and EcoRI-HF restriction digest (New England Biolabs) and ligated into pQE-60 vector using T4 DNA Ligase (NEB).

Directed library generation

A variant library was created using a modified version of Cozens' Darwin Assembly method⁴¹. Single-stranded DNA was made using Streptavidin beads and NaOH denaturation. Mutagenic primers were phosphorylated and annealed to the single-stranded DNA along with 5'-biotinylated boundary assembly oligonucleotides. A fast annealing protocol was used and this involved incubating the isothermal assembly at 95 °C for 5 min then immediately cooling to 4 °C. One volume of 2X Darwin Assembly enzyme mix was added and the reaction was incubated for 1 h at 50 °C. Streptavidin-coated paramagnetic beads were used in combination with 5'-biotinylated 5'-boundary oligonucleotides to clean up the assembly. The beads were incubated with the Darwin Assembly reaction to allow for the binding of the 5'-biotinylated boundary oligos to the streptavidin beads. After incubation, the beads were washed with NaOH to allow for denaturing of the dsDNA. Bound to the streptavidin beads was the mutated ssDNA. PCR amplification was then directly carried out using the bead-purified assembly reaction. The library fragments and the cloning vector were then restriction digested and an overnight ligation was performed.

Library transformation

Plasmid libraries were transformed into LysA *E. coli* cells (Horizon JW2806). After a 1 h recovery at 37 °C, the transformation was plated on LB plates containing carbenicillin (100 µg/mL). Transformations were scaled to achieve an average library diversity of 5×10^7 colonies per round of evolution (an average transformation efficiency of 5×10^6 colonies per transformation). Over the course of the engineering campaign, more than ten independent libraries were constructed and evaluated—including exploratory libraries not discussed in the main text—resulting in a cumulative screening total of $>5 \times 10^8$ variants. Plates were grown overnight at 30 °C. Post incubation, plates were scraped with M9 salts and the cells were pooled. Pooled libraries were centrifuged and washed with M9 salts three times at $4000 \times g$ for 10 min. Libraries were resuspended to 1 OD in M9 salts and 15% glycerol.

Genetic selection

To select for variants with activity on free CML and CML peptide, M9 minimal media agar plates (comprised of 47.8 mM Na₂HPO₄, 22.0 mM KH₂PO₄, 85.5 mM NaCl, 18.7 mM NH₄Cl, 100 µM MgCl₂, 2 mM CaCl₂, 0.4% glucose, and 7.5% agarose) were supplemented with 100–500 µM of the desired substrate and 20–80 µM IPTG, which controlled the expression of both CMLase and trypsin. LysA knockout *E. coli* (Horizon JW2806) were plated at a density to provide 10x library coverage, up to 10^8 cells per plate. A negative control without IPTG was used to verify CMLase dependent growth. All plates contained carbenicillin (100 µg/mL) to maintain selection pressure of the plasmid. Selection plates were grown at 30 °C and monitored for growth for up to one week. Once colonies were visible (1–7 days), they were picked into M9 salts and restreaked on identical selection plates to confirm IPTG dependence. Individual colonies were sequenced and mutations were identified. Colonies were then used in downstream purifications and kinetic assays.

Protein production

CMLase was expressed in LysA *E. coli* cells when they were grown in LB broth at 37 °C overnight with carbenicillin, then diluted 1:4 in Overexpression Broth (Zymo Research M3013) containing carbenicillin and 100 µM IPTG. Cells were harvested by centrifugation and lysed by a cell disruptor (PRO Scientific) at 80% amplitude and the lysate was clarified by centrifugation at 4500 rpm for 10 mins. Clarified lysate was incubated with Ni-charged Magbeads (Genscript L00295) shaking at 4 °C for 1 h. Samples were then washed twice with wash buffer (50 mM HEPES Sodium Salt pH 7.7, 300 mM NaCl, and 25 mM Imidazole) and eluted in elution buffer (50 mM HEPES Sodium Salt pH 7.7, 300 mM

NaCl, and 250 mM Imidazole). After elution, proteins were buffer-exchanged using Amicon centrifuge filters (Sigma) into protein freezing buffer (0.5x PBS, 50 mM HEPES, 15% glycerol, pH 7.0) and stored at -80 °C.

Preparation of CML-BSA

CML-BSA was prepared following previously described methods^{27,40}. Briefly, 1.2 grams of BSA was dissolved in 16 mL PBS (500 mL PSB prepared from 32 grams Na₂HPO₄ anhydrous, 2.8 grams NaH₂PO₄ - monohydrate, pH to 7.5). The following were added sequentially to 24 mL of PBS in a chemical hood: 1.8 mL NaBH₃CN (5 M stock in 1 M NaOH), 880 μ L glyoxylic acid (8.8 M stock), 400 μ L NaOH (10 M stock), 16 mL of 1.2 g BSA dissolved in PBS, filling to a final volume of 40 mL with PBS. The pH was adjusted to 8.5 and the reaction was incubated at 30 °C for 24 h. Next, the mixture was buffer exchanged into PBS to remove unreacted glyoxylic acid and sodium cyanoborohydride (30 kDa MWCO, MilliporeSigma). Protein was concentrated to ~80 mg/mL and stored at -80 °C.

Assessment of CMLase activity on CMA (carboxymethyl arginine)-BSA by LC-MS/MS

175 mg of BSA was incubated at 37 °C for 24 h in 1 mL of 0.2 M sodium phosphate (pH 7.8) containing 150 mM glyoxylic acid and 0.45 mM sodium cyanoborohydride, then dialyzed against PBS for 48 h at 4 °C. CMLase reactions were performed in 100 mM phosphate buffer at pH 8.0 with 100 μ g of glyoxalic acid-modified BSA (GA-BSA), 5 μ M of active (CrGO-897) or inactive (CeGO-763) enzyme. The total incubation volume was adjusted to 100 μ L and incubated overnight at 24 °C with shaking (300 rpm). GA-BSA incubated without the enzyme served as an additional control. For LC-MS/MS sample preparation, enzymes were added in the following order to 50 μ g of GO-BSA in 150 μ L: at 0 and 8 h, 10 μ L of 3 mg/mL protease Type XIV (Sigma-Aldrich, St. Louis, MO, Cat# P-5147); at 24 h, 4 μ L of leucine aminopeptidase suspension (Sigma-Aldrich, Cat# L5006); and at 32 h, 12 μ L of 0.5 mg/mL carboxypeptidase Y (Sigma-Aldrich, Cat# C3888). Proteolytic digestion was employed because CMA adducts proved unstable under the acid hydrolysis conditions typically used for CML-BSA processing. Before all incubations, argon was flushed after the addition of enzyme, and the entire digestion procedure was carried out in the presence of a few crystals of thymol (Sigma-Aldrich, Cat# T-0501). After 48 h, samples were centrifuged at 15,000 g for 20 min, and the supernatant was analyzed by LC-MS/MS. Phenylalanine was used as an internal standard for protein input.

Peroxidase assays

A horseradish peroxidase (HRP) reaction was performed to detect H₂O₂ levels produced by the CMLase reaction. Reactions were performed in 100 mM phosphate buffer at pH 8.0 with 0.2 U/mL HRP (Thermo J60026.MC) and 10% QuantaBlu Fluorogenic Peroxidase Substrate (Thermo 15162). Substrate concentrations varied between 10 μ M and 20 mM. Enzyme concentrations varied between 5 μ M and 0.05 μ M, with all Michaelis-Menton data at 0.5 μ M and all CML-BSA data at 5 μ M. Fluorescence was measured at 325 nm excitation and 420 nm emission and converted to H₂O₂ concentration using a standard curve measured from 5 to 30 μ M.

Enzymatic deglycation and analysis by Jess Simple Western

Deglycation reactions were performed in replicate using active (CrGO-897) and inactive (CeGO-763) enzyme variants. To evaluate enzyme efficiency, enzymes were prepared at 1 μ M. The enzymes were incubated with 5.2 ng of CML-BSA substrate. All reactions were conducted in 0.5 mL protein low-bind tubes at 24 °C with constant agitation (450 rpm) for 16 h. Following incubation, deglycation products were quantified via automated capillary electrophoresis (Jess Simple Western; Bio-Techne) according to the manufacturer's standard protocols,

with minor adjustments. Briefly, proteins were separated by size within a capillary and immobilized via UV-activated capture. Target proteins were probed using a primary anti-CML antibody (Abcam, ab27684) followed by an HRP-conjugated secondary antibody (Bio-Techne, 042-206). Chemiluminescent signals, generated by the reaction of HRP with a luminol/peroxide substrate, were detected using High Dynamic Range (HDR) imaging. Light intensity was quantified and analyzed using Compass SW software. For standard detection of CML-BSA, capillary lanes were loaded with 125 pg of substrate and incubated with anti-CML (1:200) and secondary (1:20) antibodies.

CML-modified protein ELISAs

CML-modified protein ELISAs were performed using non-specific protein-binding plates (Greiner 655061). Proteins of interest, dissolved at 0.1–1 mg/mL in 1x PBS, were added to the wells at 100 μ L per well and incubated overnight at 25 °C to facilitate binding. Following protein binding, plates were washed once with 300 μ L of wash buffer (1x PBS) to remove unbound protein. Glyoxylic acid (GA) was prepared in PBS at final concentrations of 0.5, 0.3, 0.1 or 0 mM with 1 mM sodium cyanoborohydride and added to each well (100 μ L per well). The plates were incubated at 25 °C for 1 h to facilitate the chemical modification of proteins. After incubation, wells were washed twice with 300 μ L of wash buffer. Plates were either stored at 4 °C in PBS for up to 72 h or immediately used for enzyme treatments or ELISA assays.

For ELISA assays, wells were first washed twice with 300 μ L of wash buffer. Enzyme treatment was carried out by adding 100 μ L of a reaction mixture containing the enzyme (0.05–5 μ M), 2% BSA (to maintain blocking), and reaction buffer (adjusted to pH 8). Plates were incubated at 25 °C or the desired reaction temperature for 3–18 h, with a standard incubation time of 12 h. Immediately following enzyme treatment, wells were washed once with 300 μ L PBST (1x PBS + 0.05% Tween-20).

Plates were then blocked by adding 300 μ L of blocking buffer (PBS + 2% BSA) and incubating at 4 °C for 2 h. After blocking, plates were washed three times with 300 μ L PBST. Primary anti-CML rabbit antibody (abcam27684) specific to CML was prepared at a 1:2000 dilution in PBST, and 100 μ L was added to each well. Plates were covered and incubated for 1 h at room temperature. For assays requiring secondary antibodies, wells were washed three times with 300 μ L PBST, and 100 μ L of secondary anti-rabbit antibody conjugated with HRP (abcam6721) diluted 1:2000 in PBST was added. Plates were covered and incubated for an additional hour at room temperature. Following incubation with secondary antibodies, wells were washed six times with 300 μ L PBST, with each wash lasting 5 min.

Detection was performed by adding 100 μ L of substrate solution (e.g., 1-Step Ultra-TMB) to each well and incubated for 10 min before reading absorbance at 370 nm. This protocol provided robust quantification of CML modifications and facilitated the evaluation of enzymatic activity on CML-modified proteins.

BSA deglycation and mass spectrometry sample preparation

Enzymatic deglycation reactions were evaluated for CrGO-865 and CeGO-763 (inactive control). All reactions were performed in triplicates in protein low-bind plates (Eppendorf, 003012952). Either 250, 500, 1000, or 2000 ng CML-modified BSA in 25 μ L reaction volume was incubated with enzyme samples at 5 μ M concentration for 20 h at 24 °C, with shaking at 450 rpm. Subsequently, they were reduced with dithiothreitol (DTT, 5 mM, 37 °C for 30 min), alkylated with iodoacetamide (IAA, 15 mM, room temperature, in the dark), and quenched with DTT (5 mM, room temperature, 15 min in the dark) (Thermo Fischer Scientific, A39255 and A39271). Protein cleanup was performed using a 1:1 mixture of E3:E7 Sera-Mag Carboxylate-Modified Magnetic Particles (Cytiva Life Sciences, 24152105050350 and 44152105050350) as previously described⁴² with some minor modifications. In brief, the beads were first washed with water three times, and then the sample

was added onto the beads. For protein binding, acetonitrile was added to a concentration of 75%, and the mixture was incubated for 20 min at room temperature without shaking. The beads were then washed twice with 70% ethanol, followed by once with 100% acetonitrile. The beads were resuspended in digestion buffer (PBS, pH 7.2), and then Glu-C (Promega, V1651) was added at a ratio of 1:10 (enzyme: substrate), followed by incubation for 20 h at 37 °C (1000 rpm). All digested samples were cleaned by SP3 peptide cleanup reactions in new plates due to volume constraints. In brief, 10 μ L prewashed beads were mixed with 7.5 μ L of sample, and then the solution was immediately brought to 95% isopropanol by addition of 100% isopropanol. Samples were incubated at room temperature for 20 min to enable peptide binding to the beads. The beads were washed twice with 95% isopropanol, then once with 100% acetonitrile before the peptides were eluted in two fractions using 5% acetonitrile. The cleaned up peptides were dried down in a speedvac (Labconco) and finally reconstituted in 5% formic acid, 5% acetonitrile prior to LCMS analysis.

Mass spectrometry data acquisition

All data were obtained on an Orbitrap Lumos mass spectrometer coupled to an EasyNanoLC, operating in data-dependent acquisition mode (DDA) (Thermo Fisher Scientific, San Jose, California). Peptides were separated on an Aurora Series emitter column (15 cm $\times$ 75 μ m i.d., 1.6 μ m, 120 Å pore size, C18, Elite; IonOpticks, AUR3-15075C18) using a 22 min gradient from 5 to 30% acetonitrile in 0.125% formic acid. A high-resolution MS1 scan was performed in the Orbitrap at a 120,000 resolving power, m/z range 350–1800, radio frequency (RF) lens 30%, Absolute Automatic Gain Control Value (AGC) 4×10^5 , and 100 ms maximum injection time. Ions were filtered by an intensity threshold of 5×10^4 , charge states +2–7 selected, and a dynamic exclusion was imposed which excluded ions for 15 seconds after two detection events if they occurred within 30 seconds. The parameters for MS2 acquisition were as follows: Orbitrap detector with 1.6 m/z isolation window, stepped (25, 30, 35%) normalized Higher-energy collision dissociation (HCD) energy, AGC 5×10^4 , 250 ms maximum injection time. MS2 were acquired with a resolving power of 30,000.

Mass spectrometry data analysis

Thermo BioPharma Finder 5.3 (Thermo Scientific BioPharma Finder 5.3, San Jose, California) was used to process data in peptide mapping experiments, as previously described^{43,44}, with minor adjustments. In brief, a database of known contaminants was appended with the sequence of BSA (Uniprot P02769). Variable CML-modification (+58.005 Da) was allowed on lysines, and oxidation (+15.995 Da) was allowed on methionine and tryptophan. Static carbamidomethylation (+57.021 Da) was set to cysteines. Two variable modifications were allowed per peptide. For peptide identification, all MS/MS were used, with a maximum peptide mass of 11 kDa and 5 ppm mass accuracy. Glu-C digests were set to medium specificity to allow missed cleavage peptides to be identified, and minimum confidence was set to 0. Only sites with >90% ID confidence score were kept for final analysis. A 20.0 S/N threshold was used for component detection, with all ions being used, and 5 ppm mass tolerance for peak detection. Percent abundance for a modified residue was calculated as the total peak area of components identified containing that modification, divided by the total peak area of all components containing that residue: (modified/total $\times$ 100 = % Abundance)⁴⁵. Comparisons were made across pairs at each substrate concentration. Further processing was performed in R/Rstudio (RStudio Version 2024.12.0 + 467, Posit Software) and Prism 10 (GraphPad Prism, Version 10.4.1).

Immunohistochemistry

Formalin-fixed paraffin-embedded (FFPE) tissue sections were cut at 4.0 μ m thickness. Sections were heated at 56 °C overnight to improve tissue adherence. Deparaffinization and rehydration were performed

using standard protocols. The investigator performing the staining was blinded to CMLase or inactive enzyme control status. Sections were incubated with and without enzymes, overnight (18 h) at room temperature. Subsequently, endogenous peroxidase activity was quenched by incubating sections with 10% hydrogen peroxide (H₂O₂) in distilled water for 10 min at room temperature. Sections were rinsed three times with 1X PBS for 5 min each at room temperature. Non-specific binding sites were blocked with 10% goat serum in 1X PBS for 1 h at room temperature. Following blocking, excess serum was removed, and primary antibody solution (6D12 antibody, obtained from Dr. Nagai Ryoji, diluted 1:8000) was applied to the sections and incubated overnight at 4 °C. After primary antibody incubation, sections were rinsed three times with 1X PBS for 5 min each. Sections were then incubated with biotinylated goat anti-mouse secondary antibody (from Mouse-specific HRP/DAB Detection IHC Kit, ab64259) for 15 min at room temperature, followed by three washes with 1X PBS (5 min each). Streptavidin peroxidase reagent (ab64259) was applied for 15 min at room temperature, followed by another set of three 5-minute washes in 1X PBS. DAB substrate (ab64259) was applied to the tissue for 30 seconds to develop staining, then sections were rinsed with distilled water. Counterstaining was performed using hematoxylin. Finally, sections were dehydrated through graded alcohols, mounted with Permount mounting medium, and coverslipped. Brightfield imaging of stained slides was performed using an OlyVIA VS200 scanning system under standardized settings.

Quantification of DAB staining in immunohistochemistry images

Images were loaded into QuPath for preprocessing. Full size histology image sections were processed into smaller images for quantitative analysis of DAB staining. Local features of consecutive tissue sections were visually aligned between enzyme treated and control images. Images were exported to PNG format. PNG images were loaded into Fiji (ImageJ). To separate DAB (brown) staining from hematoxylin (blue counterstain), color deconvolution was performed using the built-in “H DAB” setting in ImageJ. Briefly, images were processed via Image $\rightarrow$ Color Deconvolution $\rightarrow$ H DAB, generating three separate channels: Hematoxylin-stained nuclei (blue channel), DAB-stained structures (brown channel), and residual background noise. The DAB-only image was extracted and used for subsequent intensity quantification. Images were converted to 8-bit grayscale using Image $\rightarrow$ Type $\rightarrow$ 8-bit. To quantify DAB staining intensity, mean gray value (MGV) was measured within a region of interest (ROI) using Analyze $\rightarrow$ Measure. For each tissue section, the ROI was selected as the region of tissue positive for hematoxylin staining. The background MGV was also computed using a region of each image which was negative for hematoxylin and DAB staining. Optical density (OD) was calculated as: $-\log_{10}(\text{MGV}_{\text{ROI}}/\text{MGV}_{\text{background}})$. MGV represents the measured mean gray value of the ROI. Quantified DAB staining intensity values were exported to GraphPad Prism for statistical analysis. Data were presented as bar graphs with mean $\pm$ SEM.

Use of human tissues

Human tissue samples used in this study were obtained in accordance with the Declaration of Helsinki and approved ethical guidelines. Human tissue samples were purchased from NDRI (National Disease Research Interchange), which certifies that all samples were collected with appropriate ethical approval and donor consent.

Preparation of lens tissue and sample preparation for LC-MS/MS

A human lens was thawed on ice and homogenized with 1.5 mL of nitrogen-bubbled PBS using a hand-held glass homogenizer. The homogenate was then centrifuged at 20,000 $\times$ g for 20 min at 4 °C. The supernatant was collected and dialyzed twice with water using a 3 kDa filter to remove the buffer components and small molecules. Following

dialysis, the protein concentration was measured using the BCA assay using BSA as a standard. 100 µg of water-soluble protein was incubated with 10 µL of activated enzyme in the presence of 10 µL of the provided buffer. The total incubation volume was adjusted to 100 µL and incubated overnight at 24 °C with shaking (200 rpm). Additionally, water-soluble protein incubated in buffer without enzyme served as the control. After incubation, 50 µL of NaBH₄ was added and the mixture was incubated for 1 h at room temperature. Subsequently, 8 N HCl was added to get final 6 N HCl concentration, and the sample was transferred to a glass ampoule for overnight acid hydrolysis at 110 °C. Following hydrolysis, the sample was subjected to speed vac to remove HCl and reconstituted in 50 µL of water. After sonication and centrifugation, the sample was analyzed using LC-MS/MS. A standard addition method was performed to analyze the amount of CML modification in lens proteins.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The processed proteomics data and raw enzyme kinetics, ELISA and histology data generated in this study are provided in the Supplementary Information/Source Data file. The raw mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner⁴⁶ repository with the dataset identifier [PXD070124](https://doi.org/10.26434/chemrxiv-2024-pxd07). Source data are provided with this paper.

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Acknowledgments

We thank current and former members of the Proteomics and Proteomics Computational Teams at Calico. We also acknowledge Dr. Alessandra Zonari (OneSkin) and Dr. Hua Li (HD Biosciences) for assistance with immunohistochemistry; Dr. Vincent Monnier and Dr. John W. Baynes for helpful discussions on enzymology; and Dr. Vasiliki Chioti, Samantha Hedley, and Anika Schmidt for their contributions to developing library protocols, genetic selection techniques, and reviewing the manuscript.

Author contributions

N.T. and A.C. conceived and designed the study and computational screening. S.P. and R.N. performed CMA-BSA and lens tissue assays.

N.O., F.E.M., and J.S. designed and performed CML-BSA western blotting, proteomics, and analysis. N.T. and M.Y.H. performed library construction and screening, and conducted all biochemical characterization and kinetic assays. N.T. and A.C. oversaw immunohistochemistry and imaging analysis. N.T. and A.C. wrote the manuscript with input from all authors.

Funding

Research reported in this publication was supported by the National Institute on Aging of the National Institutes of Health under Award Numbers R43AG084351 and R44AG084351. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Competing interests

Revel Pharmaceuticals has filed a patent application on this work (U.S. Provisional Patent Application No. 64/039,597), listing N.T. and A.C. as inventors. All other authors declare no conflict of interest.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-75141-2>.

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Peer review information *Nature Communications* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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